*HS*5691 *Assignment* 2

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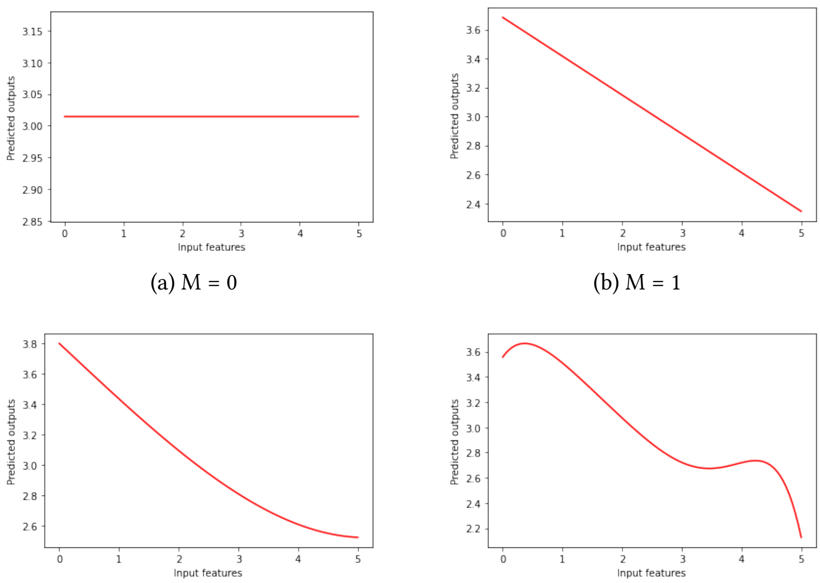
Report

1.1 Regression

In this part, we build a Least square regressor and a Ridge regressor on 1-dimensional as well as 2-dimensional dataset. We analyze our regressors for different dataset sizes, different model complexities, and different values of the regularization parameter.

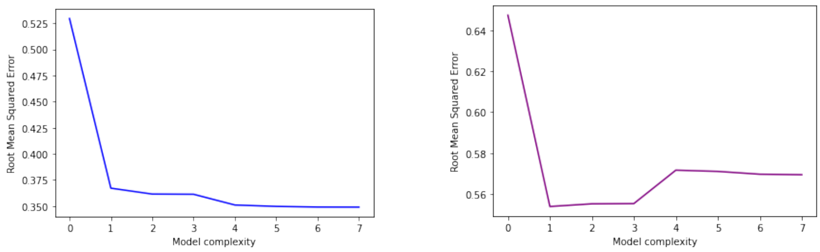
Experimental results. The optimal parameters (weights) are computed by minimizing the sum-of-squares error function for the training data. The hyperparameters (model complexity and regulariaztion parameter) were computed by plotting the E_{RMS} vs hyperparameter plot and the corresponding value of hyperparameter which corresponded to the minima of the graph was taken to be the optimal value.



(a) M = 0 (b) M = 1

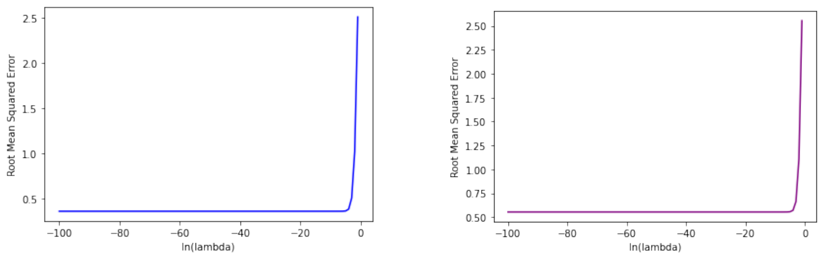
(c) M = 3 (d) M = 6

FIGURE 1.1: Plots of polynomials having various orders M



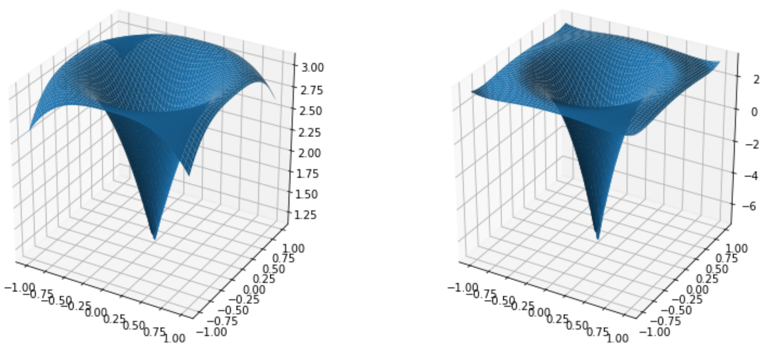
(a) E_{RMS} for training data (b) E_{RMS} for development data

FIGURE 1.2: Graphs of root-mean-squared error for varying M



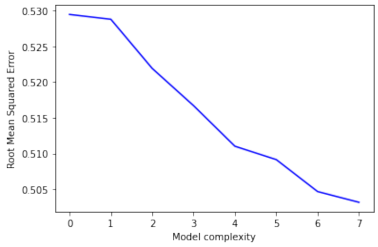
(a) E_{RMS} for training data (b) E_{RMS} for development data

FIGURE 1.3: Graphs of root-mean-squared error vs ln(λ)

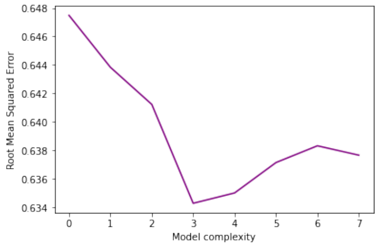


(a) Model for Least square regression for M = 3 (b) Model for Least square regression for M = 4

FIGURE 1.4: Models for varying M for 2D dataset



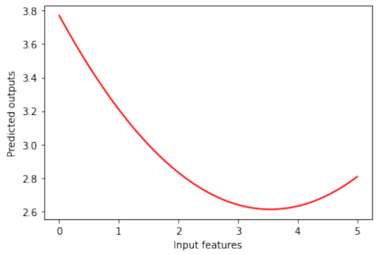
(a) E_{RMS} for training data



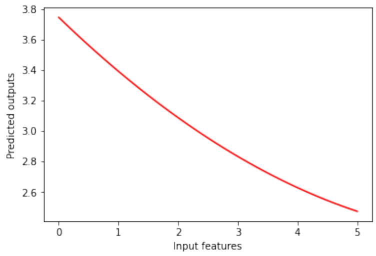
(b) E_{RMS} for development data

FIGURE 1.5: Graphs of root-mean-squared error vs model complexity for 2D least square regression

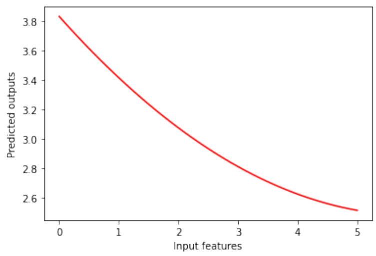
For fixed values of hyperparameters, we plotted the model functions for different number of data points.



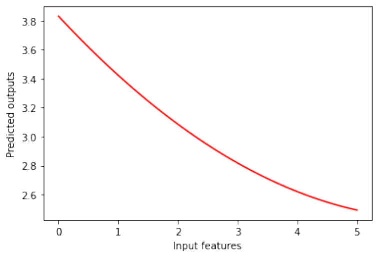
(a) N = 10



(b) N = 100

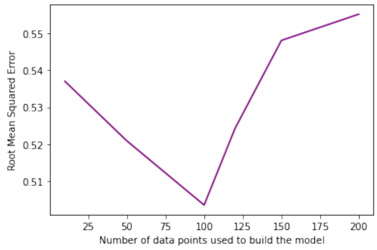


(c) N = 150



(d) N = 200

FIGURE 1.6: Plots of polynomials for N data points



(a) E_{RMS} for development data

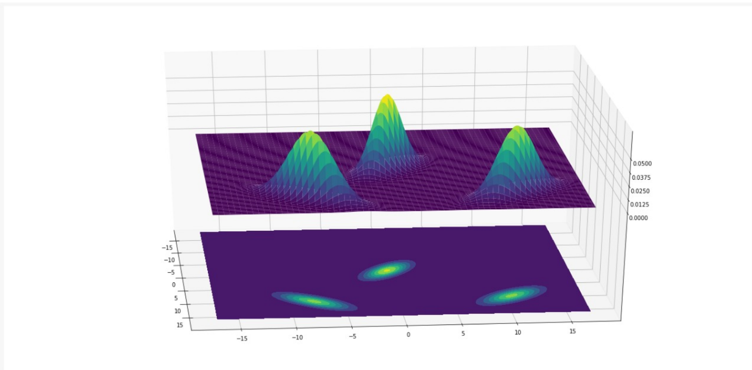
FIGURE 1.7: Graph of root-mean-squared error for varying N

Inferences. We observe from the E_{RMS} vs model complexity plot that as the model becomes more and more complex, its performance on the training data increases, but on the development data, its performance first increases but then decreases. It is clear from the theory that the issue of overfitting starts occurring.

We also observed the performance of the model by varying the number of data points (i.e, size of the training data). We observed that in general, the performance of the model increases as the number of data points increases. It is also quite evident because as our model is able to see more diverse kind of data and becomes used to it. But also, it was observed that increasing the size of training data after a certain threshold led to decline in the performance of the model. It can be inferred from the fact that sometimes redundant and more noisy data gets added to our model which leads to the decreased performance of the model.

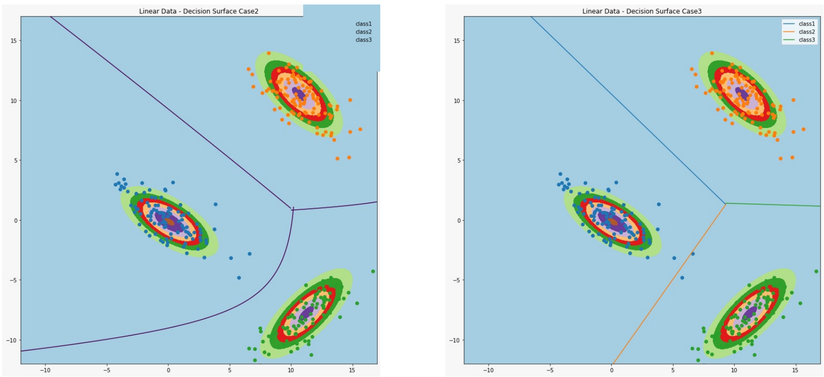
1.2 Bayesian Classifier

In this part, we build a 3-class classifier for given dataset. We assumed different cases for the covariance matrices and build the classifier accordingly. We plotted the decision boundaries and different things and also analyzed our classifier by plotting the ROC and DET curves.



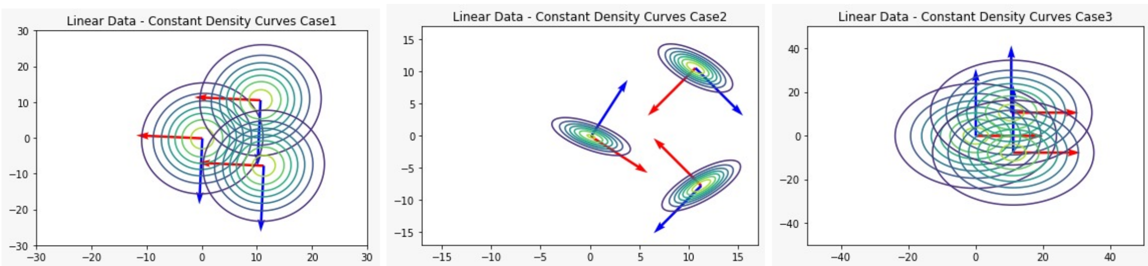
(a)

FIGURE 1.8: PDF and contours of Linearly separable Data

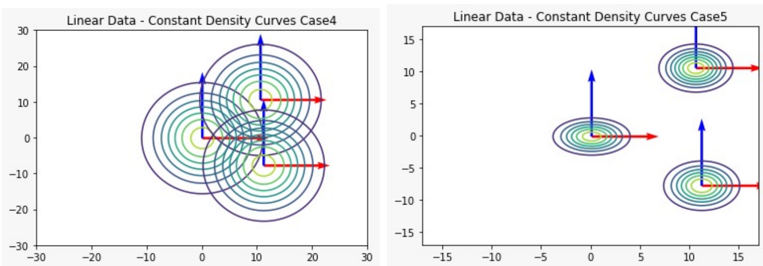


(a) (b)

FIGURE 1.9: Most general and the naive case both separate the data almost equally well for the Linearly separable Data

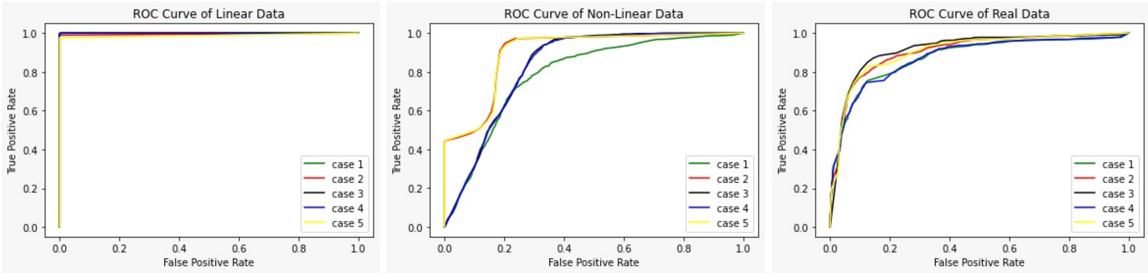


(a) (b) (c)



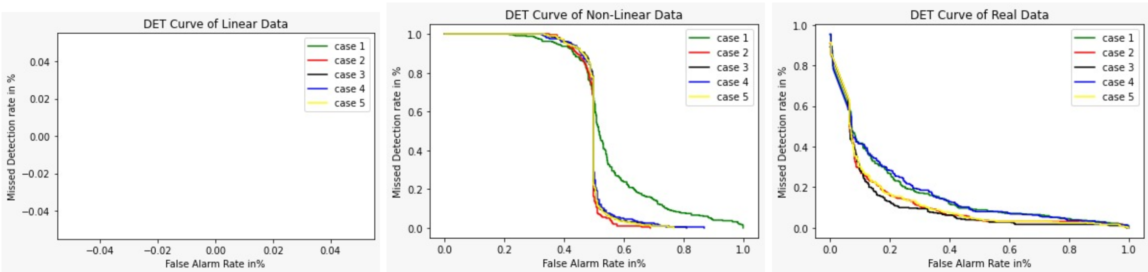
(d) (e)

FIGURE 1.10: Constant density curves. For case 1,4 and 5, we get ellipses whose major and minor axes are parallel to the coordinate axes. For case 3, we get circles becuase the covariance matrix is of the form σ^2I. For case 2, we get ellipses whose axes are not parallel to the coordinate axes

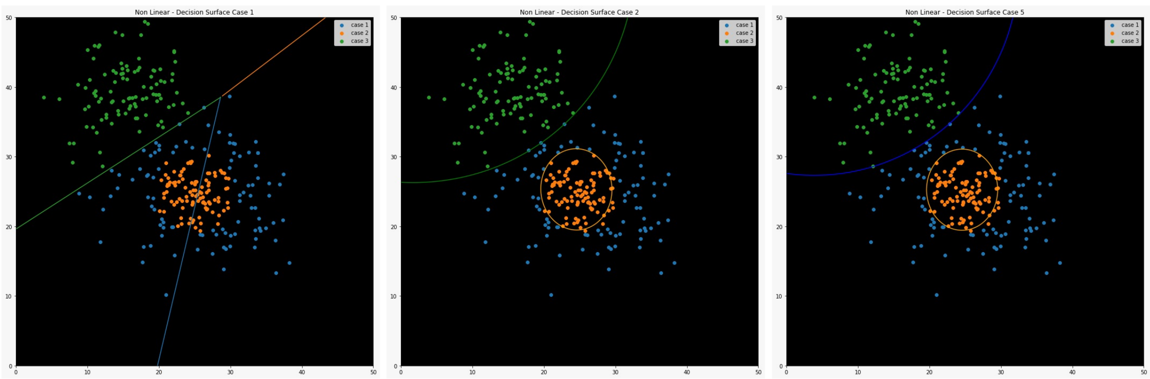


(a) (b) (c)

FIGURE 1.11: For linearly separable data, we get a highly ideal ROC curve. Case 2,5 is performing best while Case 1 is the worst performer for non-linearly separable data for and Case 3 is performing best for real separable data. Reasons for the same can be interpreted from the decision surface plots plotted below

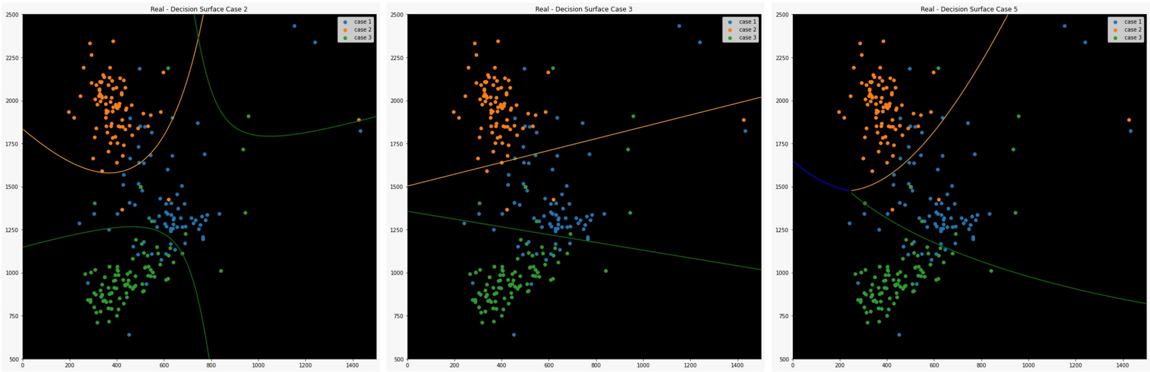


(a) (b) (c)



(d) (e) (f)

FIGURE 1.12: Case 2,5 is performing best while Case 1 is the worst performer for non-linearly separable data. Green line separates between case 1 and 3. Above part is the decision surface for case 3, below part is the decision surface for case 1, and the circle is for case 2



(a) (b) (c)

FIGURE 1.13: Case 3 performs surprisingly well followed by case 2 and 5 which we can infer from the ROC curve as well